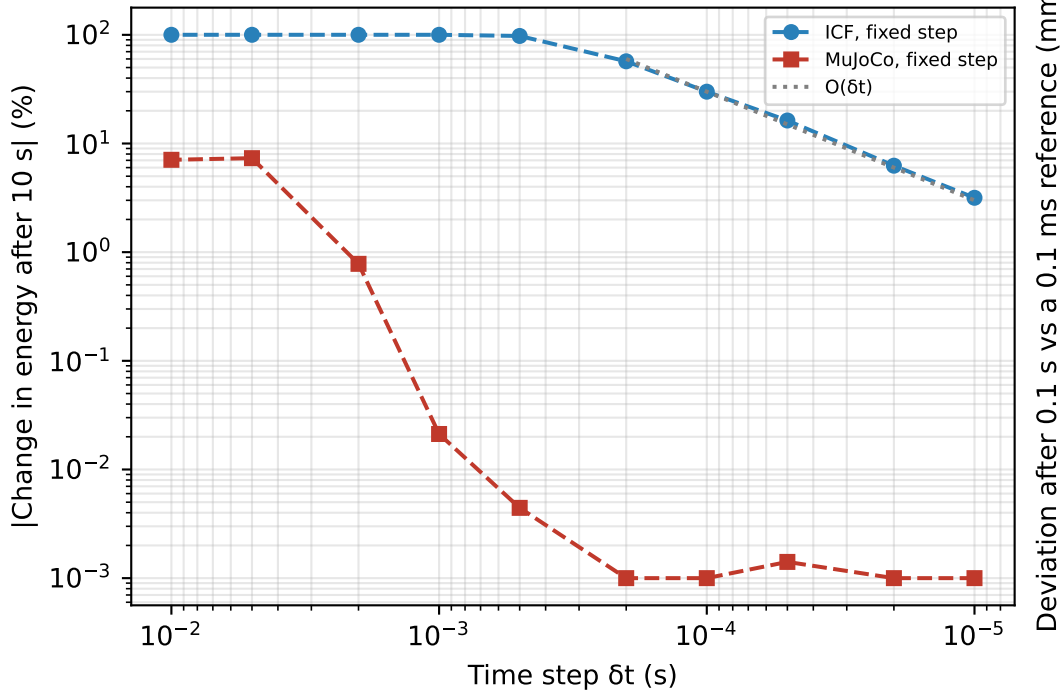


Both solvers converge; ICF realizes the model at any step, MuJoCo's constraint conserves an impact

(a) Bouncing ball, zero dissipation: ICF converges at first order; MuJoCo's undamped soft constraint holds the energy within 0.03 % at $\delta t \leq 1$ ms



(b) Soft clutter, measured error vs cost: first order for both; ICF error control $\sim 2\times$ less deviation than fixed ICF at the same cost; hollow = the reference vs itself

